# Lecture 0. Course Overview

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# Today

- Talk about the syllabus and course structure
- Assignment details
- Brief review of material from recorded first lecture
- Additional lecture with small-group problem solving

# SM615/685 - SM616/686

- This is a two-semester sequence
- How does part one relate to part two?

# SM615/685 - SM616/686

## SM615/685

- Linear Regression
- Assumptions

# SM615/685 - SM616/686

## SM616/686

- Assumptions: What happens when these break?

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For example, a key assumption of linear regression is normality:

$e_i \sim N(0, \sigma^2)$

What happens when this doesn't work?

For example, when the dependent variable is binary?

# SM615/685 - SM616/686

Another assumption is:

$Cov(e_i, e_j) = 0$ for all $i \neq j$

This is often not true. For example, in an education study which surveys classes of students.

Or when we study a time series, where observations are correlated from one time to the next.

# Prediction vs. Inference

**Inference**: describe the population, use model coefficients to explain the world.

**Prediction**: predict the value of *future* observations, model coefficients are not of direct interest, predicted values are.

# Prediction vs. Inference

## Inference Example

- What factors influence the decision to participate in a survey?
- Logistic regression coefficients are interpreted.

## Prediction Example

- Given the observed characteristics of this person and the survey design, are they likely to participate in our survey?
- The prediction is a probability.

# Syllabus

- Class starts at 9:30am
- Canvas chat is always available
- Available by appointment as well. Please contact me !

# Tools

- Course materials are adapted from James Wagner’s work
- Recorded Lectures Available on Canvas
  - Watch before class meeting!
- Class Meetings in Person
  - Meetings focus on questions, small-group problem solving

# Readings

- Faraway, *Extensions of the Linear Model in R*
- James et al. *An Introduction to Statistical Learning*
- "Class Notes” written by Steve Miller and Rick Valliant
- Other references as noted
- Find something that works for YOU

# Grading

- Homework 50%
  - R is the software we will use (If you plan to use another software, please let me know in advance)
  - Not more than week late
  - Submit via Canvas
- One Paper 20%
  - Paper 1: 4 pages
  - Paper *cannot* be late without prior permission (2 weeks prior to due date)
- Final Exam 20%
- Participation (Add notes/questions in the discussion) 10%